Coloring 101 (also known as Ramsey Theory)

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Introduction

Proposition

Among any group of 6 people, it is guaranteed that at least 3 mutually know each other, or at least 3 are mutual strangers.

Proof:

- Call them A, B, C, D, E, F.
- Out of the set $\{B, C, D, E, F\}$, A must know at least 3 people, or not know at least 3 people. Suppose WLOG that A knows B, C, D.
- Out of $\{B, C, D\}$, if any pair know each other, then that pair along with A form a group of 3. If not, then $\{B, C, D\}$ are a group of 3 mutual strangers.
- The case of A not knowing at least 3 is identical.

Ramsey Numbers

"Definition"

$R(a, b)$ is the least integer n such that when the edges of K_n are colored red and blue, there must exist a red clique of size a, or a blue clique of size b.

$R(a, b)$ is also the least n such that among any group of n people, there exists a subset of a people who mutually know each other, or a subset of b people who are mutual strangers.

We've just shown $R(3, 3) \leq 6$. To show that $R(3, 3) \geq 6$, we'd need to find a group of 5 where no 3 are mutual acquaintances or strangers.

Ramsey Numbers

Theorem (Ramsey, 1928)

$R(a, b)$ is well-defined for all positive integers a and b.

One can prove this theorem by induction in the same manner as the example.

Values of Ramsey Numbers

Now that we know Ramsey numbers exist, what are their exact values?

- $R(3,3) = 6$
- $R(4,4) = 18$
- $43 \leq R(5,5) \leq 46$
- $102 \leq R(6,6) \leq 160$
- Finding the exact value of Ramsey numbers is incredibly computationally difficult for large A, B!

A quote from Erdős:

> *Suppose aliens invade the earth and threaten to obliterate it in a year's time unless human beings can find the Ramsey number for red five and blue five. We could marshal the world's best minds and fastest computers, and within a year we could probably calculate the value. If the aliens demanded the Ramsey number for red six and blue six, however, we would have no choice but to launch a preemptive attack.*

So, what do we actually know about Ramsey numbers?

The Probabilistic Method

- Used by Erdős to show that $R(k,k)$ has an exponential lower bound.
- Non-constructively proves the existence of a coloring which avoids k-cliques for small n.
- Color the edges of K_n red or blue uniformly at random. Each of the $\binom{n}{k}$ possible k-cliques is monochromatic with probability $2^{1-\binom{k}{2}}$.
- So the expected number of monochromatic cliques is $E = \binom{n}{k}2^{1-\binom{k}{2}}$. If $E < 1$, then some coloring must have no monochromatic k-clique.
- A short computation shows that choosing $n < 2^{k/2}$ makes $E < 1$, so a k-clique-free coloring of K_n exists.
- Hence $R(k,k) > 2^{k/2}$: an exponential lower bound.

A Brief Overview of Other Ramsey-like Theorems

Van der Waerden's

Van der Waerden's Theorem

If the positive integers are r-colored, then there exist arbitrarily long monochromatic arithmetic progressions.

More on APs:

Szemerédi (1975): If A is a subset of the positive integers with positive density, then A contains arbitrarily long arithmetic progressions. This is stronger than Van der Waerden's.

Green, Tao (2004): The sequence of prime numbers contains arbitrarily long arithmetic progressions. Widely considered a major breakthrough in number theory.

Hales-Jewett Theorem

For all n and r, there exists $H(n, r)$ such that any r-coloring of an H-dimensional hypercube of side length n must contain a combinatorial line.

r players are playing tic-tac-toe on a grid of side-length n. If you make the dimension of the game sufficiently large, then a tie is impossible.

Corollary: The first player always has a winning strategy.

Finding Monochromatic Solutions

Schur (1916): If the positive integers are r-colored, then there exists a monochromatic solution to the equation $x + y = z$. The proof is a clever application of Ramsey's Theorem.

Schur's theorem, combined with some group theory, allows you to prove the following Fermat-esque result:

For all m, if p is prime and sufficiently large then $x^m + y^m = z^m$ has a nontrivial solution in the integers modulo p.

Subsequences

What's the least n such that any sequence of n distinct real numbers contains a monotone increasing subsequence of length a, or a monotone decreasing subsequence of length b?

Immediate bound: $n \leq R(a, b)$

Erdős, Szekeres (1935): $n = (a-1)(b-1)+1$

Note that we have equality here – the bound is sharp!

Erdős–Szekeres: proof via pigeonhole

Take a sequence of $n = (a-1)(b-1)+1$ distinct reals. Label each term a_i with a pair (x_i, y_i), where

- x_i is the length of the longest increasing subsequence ending at a_i, and
- y_i is the length of the longest decreasing subsequence ending at a_i.

Suppose for contradiction there is no increasing subsequence of length a and no decreasing subsequence of length b. Then $1 \leq x_i \leq a-1$ and $1 \leq y_i \leq b-1$, giving at most $(a-1)(b-1)$ possible labels.
Since there are $(a-1)(b-1)+1$ terms, by the pigeonhole principle two terms a_i, a_j with $i < j$ share a label. But if $a_i < a_j$ then $x_j > x_i$, and if $a_i > a_j$ then $y_j > y_i$ – either way their labels differ, a contradiction. ■

The Big Picture

All of these theorems are about taking a chaotic structure (r-colorings) and proving the existence of an orderly substructure (cliques, APs, lines).

Van der Waerden's and Hales-Jewett use a very similar technique of color-focusing in their proofs. It turns out that there is an injective correspondence from lines to progressions, so Hales-Jewett implies Van der Waerden's.

References

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